DATA601-25X

Coastal Erosion Study In Canterbury

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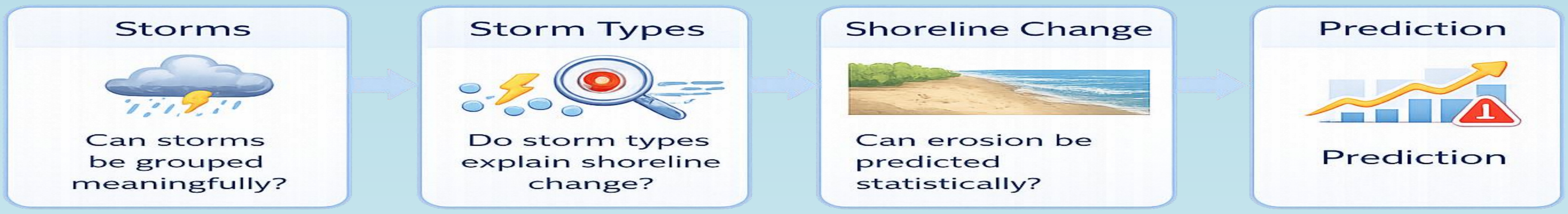
Why This Study

Coastal erosion along the Canterbury coast occurs primarily during storm events.
However, storms differ in intensity, duration, and wave direction, leading to highly variable shoreline responses.

Data

- ECAN wave buoy: Observed storm wave height, duration, direction
- NZ wave hindcast: Modelled storm wave height, duration, direction
- CoastSat: satellite-derived monthly shoreline positions

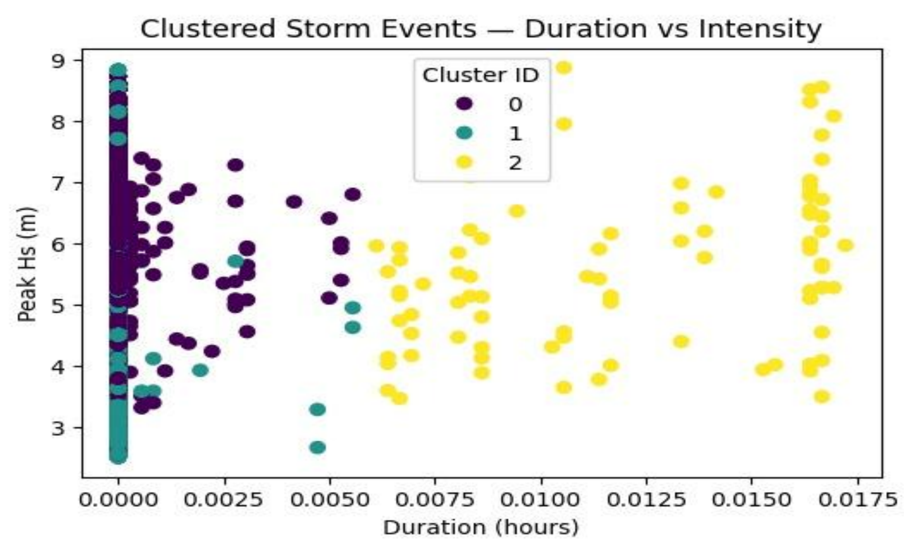
Objectives



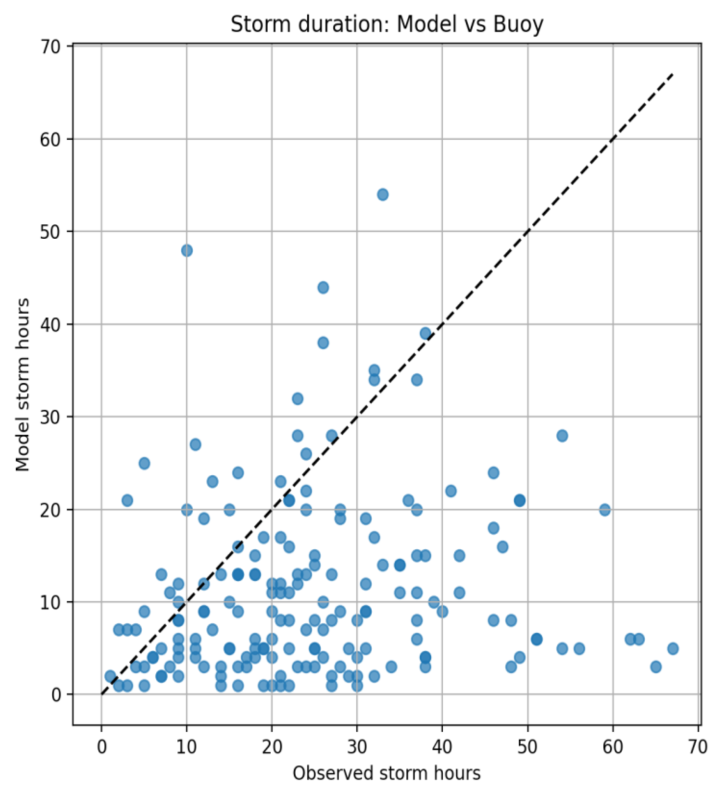
Method

Storm detection: Threshold-based identification of storm events from wave time series
Storm typology: K means clustering by intensity, duration, and direction
Validation: Hindcast storms compared with ECan buoy observations
Shoreline response: Monthly CoastSat shoreline change linked to storms
Prediction: Logistic regression of erosion probability

Results



- **Storm Type 0:** Short-duration, low to moderate waves; typically associated with little or no erosion.
- **Storm Type 1:** Moderate wave height and duration; occasionally linked to erosion under favourable conditions.
- **Storm Type 2:** Long-lasting, high-energy storms; responsible for the most severe and extreme erosion events.



- Hindcast storm duration shows increasing underestimation for longer events compared with ECAN buoy observations.

Key Findings & Conclusion

- Storms naturally group into short, weak events and long, powerful events with very different coastal impacts.
- Storms approaching from certain directions deliver more wave energy to the Canterbury coast, increasing erosion risk.
- Modelled storms do not always match buoy observations, especially for storm strength and duration, affecting erosion estimates.
- Similar storms can produce very different shoreline responses, depending on prior beach conditions and storm history.

Future Work

Incorporate nearshore wave transformation, antecedent beach state, and storm sequencing to improve erosion risk assessment.